MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, THERMOCOUPLE (FOR MIL-C-5015 MS3450 SERIES AND MIL-C-83723 SERIES II CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.

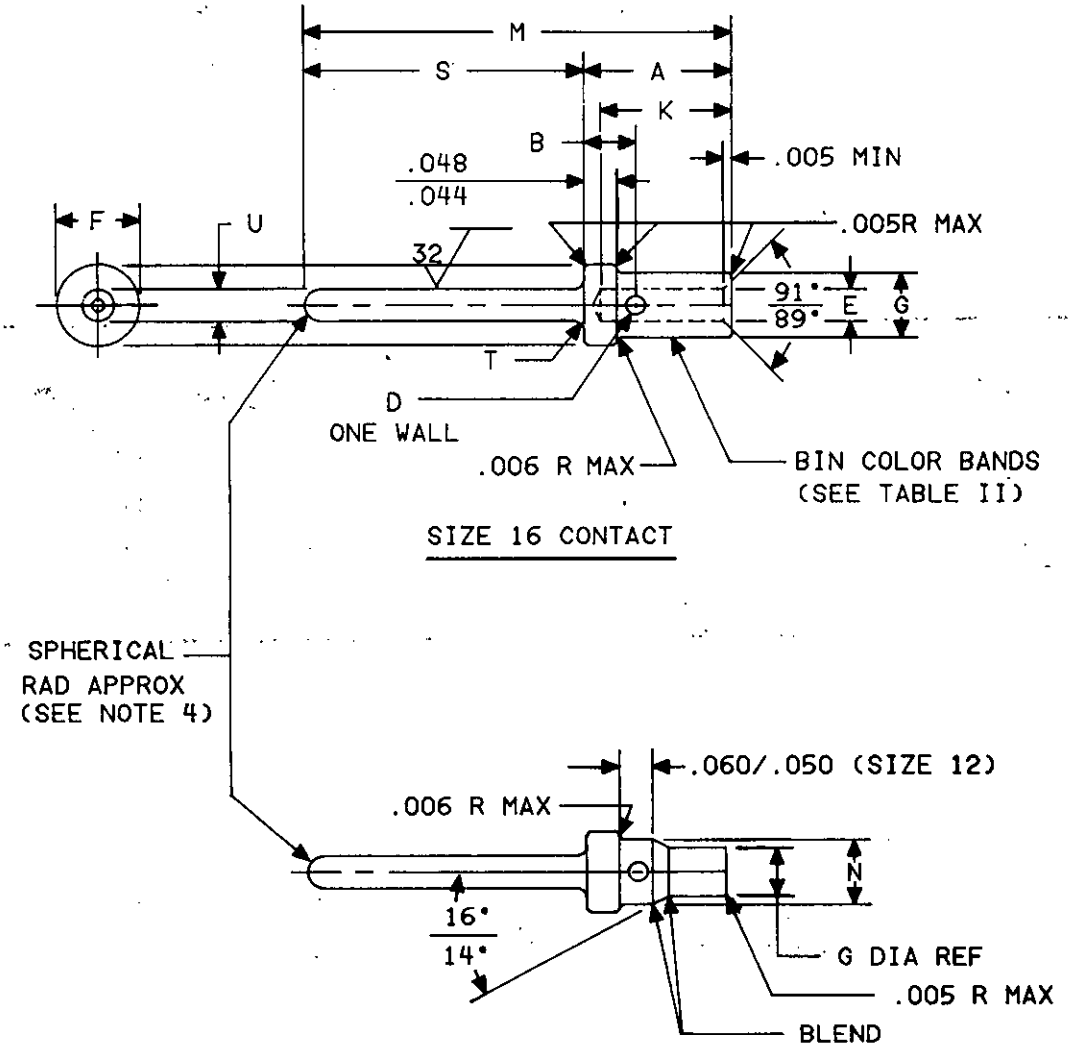


FIGURE 1. Pin contacts.

Ⓑ denotes change

FSC 5935

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INCHES	MM	INCHES	MM	INCHES	MM
.005	0.13	.073	1.85	.160	4.06
.006	0.15	.079	2.01	.186	4.72
.015	0.38	.093	2.36	.189	4.80
.036	0.91	.095	2.41	.250	6.35
.042	1.07	.098	2.49	.290	7.37
.044	1.12	.101	2.57	.300	7.62
.048	1.22	.102	2.59	.619	15.72
.050	1.27	.103	2.62	.631	16.03
.060	1.52	.130	3.30	.808	20.52
.0615	1.560	.133	3.38	.819	20.80
.0635	1.613	.147	3.73	.923	23.44
.066	1.68	.150	3.81	1.110	28.19
.068	1.73	.155	4.01		

TABLE I. Dimensions.

BIN code	A	B	D DIA	E DIA	F DIA	G DIA	K	M (REF)	N DIA	S	T MAX	U DIA
454 thru 457	.300	.079	.042	.068	.133	.103	.250	.923		.631	.015	.0635
				.066	.130	.101				.619		.0615
458 thru 461	.290	.073	.036	.102	.189	.150		1.110	.160	.819		.095
				.098	.186	.147			.155	.808		.093

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Spherical end dimension shall conform to MIL-C-39029.

FIGURE 1. Pin contacts - Continued.

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REQUIREMENTS:

Dimensions, design characteristics, and configuration: See figure 1 and tables I and II.

Mating contacts: MIL-C-39029/86

TABLE II. Design characteristics.

Bin Code	1st	2nd	3rd	Mating end size	Wire barrel size	Type	Class
454	Yellow	Green	Yellow	16	16	(See table IV)	B
455	Yellow	Green	Green	16	16		
456	Yellow	Green	Blue	16	16		
457	Yellow	Green	Violet	16	16		
458	Yellow	Green	Gray	12	12		
459	Yellow	Green	White	12	12		
460	Yellow	Blue	Black	12	12		
461	Yellow	Blue	Brown	12	12		

Tools: See table III.

TABLE III. Tools.

BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool
454 thru 457	M22520/1-01 thru M22520/7-01	M22520/1-02 Blue M22520/7-03	M81969/14-03	M81969/14-03
458 thru 461	M22520/1-01	M22520/1-02 Yellow	M81969/14-04	M81969/14-04

Material, material code, and plating: See table IV.

Tensile strength: See table IV.

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TABLE IV. Materials, plating, and tensile strength.

BIN code	Material	Plating	Tensile strength 1/					
			Wire size	Axial load	Wire size	Axial load	Wire size	Axial load
454	Constantan	None	20	45 lbf	18	95 lbf	16	110 lbf
455	Alumel	None	20	60 lbf	18	110 lbf	16	125 lbf
456	Chromel	None	20	70 lbf	18	125 lbf	16	145 lbf
457	Iron	Cadmium plate 2/	20	45 lbf	18	95 lbf	16	110 lbf
458	Constantan	None	---	---	14	210 lbf	12	235 lbf
459	Alumel	None	---	---	14	210 lbf	12	235 lbf
460	Chromel	None	---	---	14	240 lbf	12	265 lbf
461	Iron	Cadmium plate 2/	---	---	14	210 lbf	12	235 lbf

1/ Applies to type 1 wire per MIL-W-5846.
2/ Chromate clear coat.

Vibration: Test condition VI, letter J per method 2005 of MIL-STD-1344, except duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction for a total of 16 hours.

Shock (specified pulse): Test condition D per method 2004 of MIL-STD-1344, except the pulse shall be an approximate half sine wave of 300 G ±15 percent magnitude with a duration of 3 milliseconds ±1.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

Military part number: See table V.

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TABLE V. Military part number and BIN code.

BIN code	Military part number	Supersedes
454	M39029/85-454	M39029/851616C1
455	M39029/85-455	M39029/851616C2
456	M39029/85-456	M39029/851616C3
457	M39029/85-457	M39029/851616C4
458	M39029/85-458	M39029/851212C1
459	M39029/85-459	M39029/851212C2
460	M39029/85-460	M39029/851212C3
461	M39029/85-461	M39029/851212C4

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC
Air Force - 11, 15, 99
DLA - ES

User activities:
Army -
Navy - OS, SH

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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